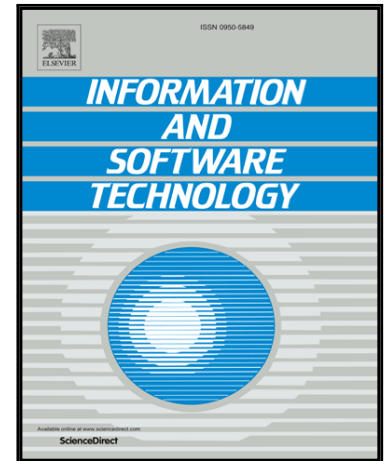


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Applying Process Mining Techniques in Software Process Appraisals

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Abstract

Context: Process assessments are performed to identify the current maturity of organizations in relation to best practices. Existing process assessment methods, although widely used, have limitations such as: dependence on the competencies of appraisers; high amount of effort and resources required; subjectivity to analyze data and to judge on the implementation of practices; low confidence in sampling and its representativeness. Currently, due to the increasing use of information systems to support process execution, detailed information on the implementation of processes are recorded as event logs, transaction logs, etc. This fact enables the usage of process mining techniques as a powerful tool for process analysis. It allows using a significant amount of data with agility and reliability for process assessments.

Objective: The objective of this paper is to present the development and application of a feasible, usable and useful method, which reduces the limitations of current SCAMPI method and defines how to apply process mining techniques in SCAMPI-based process assessments

Method: Research method comprises nine steps that were performed in a manner that raised questions in the first four steps were answered by the last four steps of the research design.

Results: The method “Process Mining Extension to SCAMPI” was designed, developed, applied in two cases and submitted for review by experts who judged it viable, usable, and useful.

Conclusions: As per this research, process mining techniques are suitable to be applied in software process assessments since they are aligned with the purposes of data collection and analysis tasks. In addition to that, the resulting method was judged by experts as something that reduces identified limitations of one of the most used process assessment method.

Keywords: Process mining; Software process assessment; SCAMPI; Data collection and analysis; Process Mining Extension to SCAMPI.

1. Introduction

The software development process is widely recognized as a key factor that contributes to the quality of software. To systematize software development processes, reference models are considered, since they provide a set of universally accepted best practices used as a reference for the creation of such processes. Differences between the actually performed processes and the documented processes have already been recognized by the research community [1]. In this context, the challenge is to determine how much a software process, as performed, is in accordance with the reference model. This can be done through software process assessments. Rout et al. [2] referred to process assessment as the disciplined examination of the process used by an organization in relation to a set of criteria to determine the ability of these processes to run within quality, cost and schedule goals. One of the best-known process assessment methods is the Standard CMMI Appraisal Method for Process Improvement (SCAMPI) v1.3b [3]. It is used to identify strengths, weaknesses and ratings related to the Capability Maturity Model Integration (CMMI) reference models [4].

Despite the fact that SCAMPI method is mature and, along with a CMMI model, it has been used to assess thousands of organizations worldwide [5]. Chen et al. [6], Northcutt and Paulk [7] and Margarido et al. [8] mention that process software assessments have drawbacks that typically make its execution time and resource consuming, especially in large organizations. As noted by to Chen et al. [6], process assessments are generally manual, which usually makes a simple evaluation an inefficient process; external appraisers are frequently not allowed to directly access the information due to security and privacy concerns (i.e. restriction of authority) and assessments are based on subjective evaluation, causing bias results and usually require experienced appraisers to understand specific software processes and to prepare relevant questions for interviews.

Although there are some initiatives to create “lightweight” process assessment (and improvement) methods such as Adept [9], iFLAP [10], METvalCOMPETISOFT [11] and the ones (e.g [12]) based on ISO/IEC TR 29110, studies such as Rout et al. [2] point out that the costliest activity of an appraisal is evidence collection, representing 47% of total effort. Moreover, this fact is aggravated since in an appraisal, typically only a small sample (typically less than 25% of all process instances, as per one of the author’s practical experience) of the process execution is considered. It is corroborated by Northcutt and Paulk [7] who stated that process assessments typically deal with populations “*in the tens and similarly small sample sizes*”. In addition, data collection and analysis techniques that consider the recent capabilities in information technology, such as data, text and process mining are not employed. The idea of process mining is to discover, monitor and improve real processes by extracting knowledge from event logs that are readily available in modern information systems [13]. Process

mining can answer questions such as "how much the actual process deviates from the official process?". Samalikova [1] points out that during an assessment, similar questions are made and need to be answered, and therefore process mining can be recognized as a valuable mean to collect the objective evidence necessary for process assessments.

Process mining allows a comparison between how the processes actually run against the way they were designed to operate. Aalst [14], Samalikova [1], Riera Cruaños [15], Rubin et al. [16] and Samalikova et al. [17] suggest the promising use of process mining. The premise is that mature companies have been effective in collecting, organizing and storing a large amount of data from their daily operations. Most of these companies, however, do not use these data correctly, so as to turn them into knowledge to be employed in process assessment activities. So, there is an opportunity of applying processes mining techniques as an additional approach to the data collection and analysis techniques already existing in SCAMPI v1.3b method.

The main objective of this paper is to present the development and application of a feasible, usable and useful method, which reduces the limitations of current SCAMPI method and defines how to apply process mining techniques in SCAMPI-based process assessments.

The remaining part of the paper proceeds as follows: section 2 presents the research design, section 3 describes the background and related work; section 4 and 5 show the development and application of the method, respectively; section 6 presents the discussion and interpretation of results. Finally, section 7 ends the paper with conclusions.

2. Research design

As the research design, the method development and application were conducted through nine methodological steps, as presented in the Fig. 1. This methodological approach was inspired on a development lifecycle developed in 1991 by NASA, the V-model [18], where elements in the right-hand side aim to answer questions presented in steps in the left-hand side. Moreover, steps were identified to support the development of the method similarly to the conduction of engineering-like tasks, such as requirements definition, design, verification & validation, etc. In the first step, the research problem is defined. Research objectives and method are outlined in the second step. Subsequently, a literature review is performed in step 3. After, in step 4- Development of "Process Mining Extension to SCAMPI", requirements for the extended method are identified and implemented, resulting in the extended method itself. In order to test it, a running example (i.e. a simulation of the application of the method) is conducted in step 5. Verification and validation of the method are conducted in step 6 and 7, respectively via cases study and a review done by CMMI and SCAMPI experts, similar to the expert panel proposed in Beecham et al. [19]. The cases study, apart from being the mechanism to entirely test the method, also aims to check whether the requirements for the method as well as its adequacy of use are met. The review by experts aims to check whether limitations identified in step 3 are properly addressed by the extended method. Step 8 covers the interpretation and discussion of results. It also includes checking if research objective is achieved. Finally, step 9 presents the conclusion, including checking whether research problem is solved).

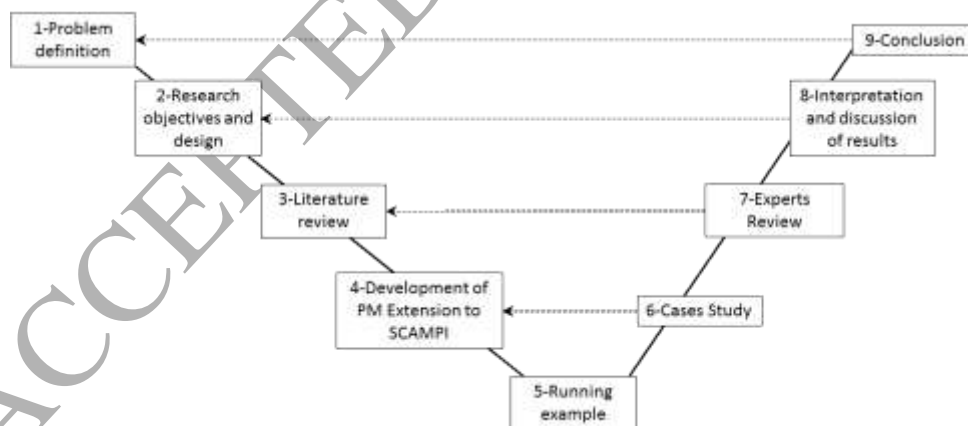


Fig. 1. Research Design

As part of step 1, the following research problem was defined:

"The SCAMPI method presents limitations, such as the dependence on the competencies of appraisers; low cost-benefit ratio; subjectivity for data analysis and judgment regarding the implementation of practices; and low confidence in the selection of the sample and its representativeness. Process mining can be applied to aid collecting and analyzing data in

SCAMPI appraisals. However, there is a lack of a method that defines what, when, where, how and why to apply process mining techniques in software process appraisals, aiming to reduce the limitations of the current SCAMPI method."

Derived from such research problem, the following research objective was stated in step 2:

"Develop a feasible, usable and useful method that reduces the limitations of the current SCAMPI method and defines which, when, where, how and why to apply process mining techniques in SCAMPI-based process assessments".

3. Background and related work

This section covers topics related to process assessment methods and process mining. The exposition of such aspects are important to understand the concepts, principles as well as the motivation for the work presented here, which integrates these two important topics. The section also presents related work.

3.1. Process assessment methods

Process assessment has been defined, in [4], as the examination of one or more processes by a team of trained professionals, using a reference model as a basis for determining, at a minimum, strengths and weaknesses. An assessment method comprises the activities that must be performed in order to carry out a process assessment. Characteristics of assessment methods can vary, including duration and effort, appraisal team composition and depth of investigation [4]. SCAMPI v1.3b is an example of a process assessment method. It is used to identify strengths and weaknesses in relation to CMMI reference models. It is adherent to ARC-Appraisal Requirements for CMMI v1.3 [20] and it consists of several processes and activities divided into four phases: 1-plan and prepare for appraisal, 2-conduct appraisal, 3-report results, and 4-action plan reappraisal (optional).

According to the SCAMPI method, process appraisals depend on an aggregation of information that is collected via defined types of objective evidence. Objective evidence refers to artifacts or statements used as indicators of the implementation and institutionalization of model practices [3]. For instance, the risk register for a project is an example of objective evidence. The extent to which an objective evidence is considered adequate to determine that a part of a given model element is implemented will vary according to the context in which the process is applied, and influenced by factors such as size, organizational culture, application domain, market and so on [3]. In general, SCAMPI method typically use approaches as questionnaires, document review, interviews and demonstrations to collect information about processes in an organization. In particular, data collection and analysis are performed using the following types of objective evidence:

- Artifacts: tangible objective evidence, indicative of the work being done, which is the primary output or a consequence of the implementation of a model practice [3]. To verify the implementation of practices, it is necessary sufficient artifacts that demonstrate and confirm that the work is being done. Such verification is done by appraisers who review the artifacts resulting from the process implementation;
- Affirmations: oral or written statements, provided by the practitioners, which substantiate the implementation of model practices [3]. Affirmations are usually collected through a variety of techniques, including interviews, presentations, demonstrations, surveys or other means.

3.2. Process Mining

According to [14], process mining is defined as a technique to discover, monitor and improve real processes by extracting knowledge from event logs readily available in information systems. Events may be stored in database tables, message logs, mail archives, transaction logs, and other data sources. It addresses the problem that most businesses have very limited knowledge about what is actually happening in their organization. It can be considered as proficient means for helping organizations understand their actual way of working and can serve as a foundation for process improvement. The goal of process mining is the discovering, monitoring and improving of processes by extracting knowledge from their execution traces [21]. Based on the event traces, models that describe underlying processes can be extracted.

As illustrated in Fig. 2, there are three main types of process mining: discovery, conformance and enhancement.

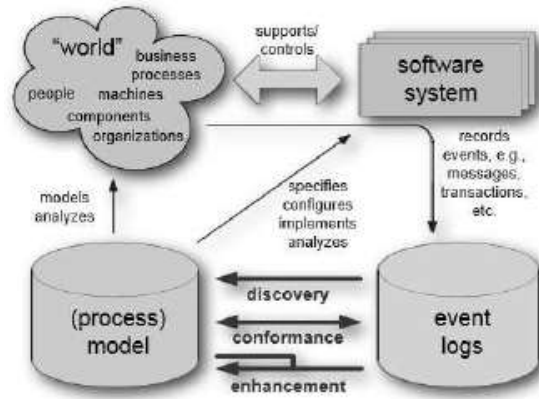


Fig. 2. Main types of process mining, reproduced from [14]

The objective of discovery process mining is to discover process models based on event logs. For example, the alpha-algorithm [22] takes an event log as its input and automatically constructs a Petri net to explain the behavior recorded in that log.

The conformance process mining aims to verify an actual process through performing the comparison between its existing process model and the event log of that process, and vice versa [23][24]. For example, a conformance algorithm is proposed by Rozinat and Aalst [23] to quantify and diagnose deviations between a given model and its corresponding event log. Essentially, conformance is to ensure that business processes, operations and practices conform to a set of prescribed or agreed rules [25]. Conformance checking techniques examine the degree to which the process execution corresponds to a particular normative model process. In addition, conformance checking can point out the parts of the process that the log does not comply. Other relevant conformance checking technique is the LTL Checking [26], which analyzes the log with regards to conformity to specific restrictions, specified by linear temporal logic formulas. This technique can be used where there is not a complete process model a priori but only a set of requirements, such as business rules. According to [27], despite their importance, few conformance checking algorithms are present in the literature.

While the conformance checking determines the positioning between the model and reality, the third type of process mining, called enhancement (or extension), aims to improve or enhance the process model based on information extracted from a log.

Van der Aalst [14] argues that there are two types of process models, *de Jure* and *de Facto*. *De Jure* models specify the expected behavior for a process, or, in other words, how things should be done while *de Facto* models reflect the actual process being executed. Van der Aalst et al. [28] proposed a framework called "Audit 2.0". One component of this framework is the conformance checker. It checks if data relating to the process execution are in accordance with the *de Jure* models and *de Jure* business rules. This includes the process behavior (i.e. control-flow) as well as the data flow, performers and business rules [28].

3.3. Related work

Some papers already proposed the application of process mining in the software process context.

Rubin et al. [16] developed a process mining framework for software processes where different aspects of process mining were discussed, such as control-flow perspective (that captures the order in which activities are performed), the cases perspective (capturing data, documents and information required or produced in one case) and organizational perspective (identifying people or roles that perform a particular activity). Their work presents some algorithms such as Alpha algorithm, Multi-phase miner, Heuristics miner, Conformance checking algorithm, LTL Checker, Social network miner and Organizational miner. However, the authors conclude that there is a lack of algorithms to generate formal models for software processes.

Samalikova developed her doctoral dissertation [1] that promotes the use of process mining for appraising and improving software processes from suitability evaluation of use and the definition of approaches to do this. The conclusion was that information collection techniques derived from software process mining can be applied to improve information collection in software process assessments. However, the scope of investigation was restricted to generic practices of CMML. Also, process mining types, techniques and procedures used were not nominally indicated. Additionally, Samalikova et al. [17] point out that, in software development environments, records obtained from software development supporting tools, such as configuration management system, development tools, etc, are used to generate the process model being practiced. This

model can be analyzed, verified, optimized and then managed. In other words, process mining can be used not only to discover but also to monitor and improve real software processes using data obtained from repositories.

Riera Cruanas developed his master thesis [15] to investigate the literature on supporting tool for CMMI for Development (CMMI-DEV) key process areas to find out whether the use of process mining is possible for improving the assessment of these process areas and, by extension, the assessment of the CMMI model in general. However, objective criteria for such judgment were not developed and no particular process mining technique was pointed out.

Lemos et al. [29] intended to complete the gap between process mining and software engineering through demonstration of how to explore process mining techniques to conduct conformance analysis and assess whether the software development process has been followed. The authors focused on control-flow perspective by applying the sequence clustering analysis process mining algorithm on a real event log of a software house. Although valid, the study was restricted to a single algorithm without establishing a guide for other algorithms, or considering other perspectives.

As a result of the systematic literature review conducted in this research, it can be said that although there are papers that describe cases of the application of process mining in software process assessments, this is still incipient, and clearly there is a gap in terms of a formal method to guide appraisers on what, how, where, when, and why to apply process mining techniques in SCAMPI appraisals.

4. Method development

The development of the method is described in this section. However, before describing it, the concept behind the proposed method needs to be presented.

4.1. Overview of the extended method

CMMI models are collections of best practices that help organizations to manage and improve their processes [4]. Currently, SCAMPI method is used to identify strengths and weaknesses relative to a CMMI model. It incorporates recognized best practices from the appraisal community, and is based on the features of several legacy appraisal methods. Fig. 3 shows how Process Mining discipline, CMMI model and SCAMPI method relate to each other and also to the proposed method. The purpose of “Process Mining Extension to SCAMPI” is to provide an extended appraisal method capable of supporting process mining-aided appraisals, where process mining techniques, such as process discovery and conformance checking, support data collection and analysis [30]. So, the idea is to guide CMMI appraisers on what, how, when, where and why apply Process Mining techniques on CMMI-based appraisals. “Process Mining Extension to SCAMPI” does not intend to replace SCAMPI Method Definition Document (MDD) v1.3b [3] since it does not replicate its full content. It is positioned as a set of new or extended processes, activities and its derived elements (e.g. inputs, outputs, tools and techniques) in SCAMPI.

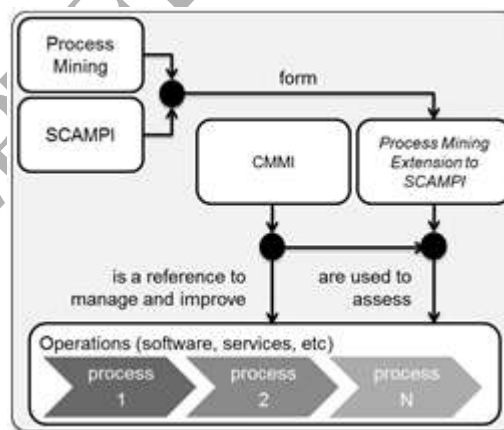


Fig. 3. Process Mining Extension to SCAMPI, reproduced from [31]

“Process Mining Extension to SCAMPI” [31] has been designed to reduce specific SCAMPI limitations, which were identified via process assessment community feedback. Requirements that address such limitations were used as references for key method architecture and design decisions. “Process Mining Extension to SCAMPI” intends to be utilized in the same context as SCAMPI (i.e. internal process improvement, supplier selection, process monitoring) as long as there are pre-conditions (i.e. premises) to apply Process Mining techniques, as stated in the Process Mining Manifesto [32]:

- Possibility to sequentially record, from information systems that support process execution, events in a manner that each event refers to an activity and is related to a particular case (i.e., a process instance);
- Whenever possible, record extra information such as the resource (i.e., person or device) executing or initiating the activity, the timestamp of the event, and data attributes related the event (e.g., the number of errors found in testing).

4.2. Development of “Process Mining Extension to SCAMPI”

In order to develop the proposed method in a structured way, specific goals of Service System Development (SSD) process area of CMMI for Services v1.3 [33] were considered as a reference to determine tasks to develop and apply the method. Thus, the development and application of the extended method was performed through the following ten tasks:

4.2.1 Identify and analyze needs, expectations and constraints for the proposed method

In this task, papers presenting limitations of process assessments were analyzed. Field experience from one of the authors was also taken into consideration. A list of limitations was identified and, in order to get corroboration of the limitations, an online questionnaire was constructed in Poll daddy tool (www.poll daddy.com) and applied among appraisers and other CMMI experts, through specific forum of such “community” at LinkedIn (www.linkedin.com). Such questionnaire was entirely answered, using a continuous scale from 1 to 5, by 25 persons. Six respondents answered it incompletely, and thus, the completion rate was 80.6% and Cronbach's alpha was 0.8534, representing a good internal consistency.

The requirements for the extended method were also identified in this task. Requirements state what the method needs to address to reduce identified limitations. In Table 1, limitations and their derived requirements can be seen.

Table 1 - Limitations and requirements for the extended method

Limitation	Requirement
SCAMPI appraisals are inherently dependent on the appraisers and the competencies of appraisers.	Extended method should reduce the dependency on appraiser and their competencies.
SCAMPI appraisals are long, complex, expensive and resource demanding (especially regarding data collection and analysis tasks).	Extended method should reduce the amount of effort and time from appraisal team (especially regarding data collection and analysis tasks).
SCAMPI presents subjectivity to analyze data and to judge about the implementation of practices.	Extended method should reduce subjectivity to analyze data and to judge about the implementation of model practices.
SCAMPI does not provide confidence regarding sample selection and its representativeness.	Extended method should increase the confidence regarding sample selection and its representativeness.

4.2.2 Characterize the approaches for collecting and analyzing information on current SCAMPI method

In this task, data collection and analysis techniques in the SCAMPI method were investigated, in order to define how they would be addressed in the extended method. SCAMPI method requests a data collection strategy which comprises the data collection approach (discovery, managed discovery or verification); when data will be collected; which techniques (documentation review, demonstrations, presentations, interviews and questionnaires) will be employed to both objective evidence types: artifacts and affirmations; responsibilities for data collection; etc. In addition, a data collection plan, derived from data collection strategy, should also be established [3]. Typically, it contains:

- register of data collection strategy;
- identification of participants for interviews;
- process areas assignment for team members;
- readiness review schedule and criteria;
- summary of initial objective evidence provided by the organization;
- prioritized data needed;
- schedule of affirmation collection activities;
- initial questions for interviews;
- needed artifacts identification;
- risks related to data insufficiency and inappropriate schedule.

In order to maintain the adherence to ARC (as original SCAMPI method does), it was noted that process mining techniques should be aligned to the characteristics of approaches regarding collecting and analyzing information on the SCAMPI method. So, data collection strategy as well as its data collection plan should describe how process mining is

applied in a SCAMPI assessment. In addition, process mining, due to its particular characteristic of using electronically documented records (as opposed to verbal affirmations), would be applied in relation to artifact-type objective evidence. In terms of data collection approach, process mining is exclusively associated with the “verification” approach described in SCAMPI method.

4.2.3 Characterize the existing process mining techniques, identifying which can be used in SCAMPI appraisals

It was identified that process mining, when applied in the context of SCAMPI appraisals, primarily would use the process mining techniques called “process discovery” and “conformance checking” and the algorithms associated with such techniques. The first is due to the fact that in a SCAMPI appraisal, the process actually being performed is evaluated and captured by this technique. Conformance checking (and derived algorithms) would be used since it is the main technique to verify, for example, whether the process being performed follows CMMI practices or the procedure defined by the organization. Business rules conformance checking would also be used. In Table 2, some of the most relevant process mining algorithms and their outputs for each process mining technique are presented.

Table 2. Process mining techniques and related algorithms

Technique	Purpose	Algorithm	Output
Process Discovery	To discover the actual process being performed (i.e. <i>de Facto</i> Model).	Alpha miner; Alpha++	Petri Net
		miner; ILP miner	
		Evolutionary Tree Miner	Process Tree
		Inductive Miner	Petri Net or Process Tree
		Fuzzy miner	Fuzzy Model
		Genetic Miner; Heuristic miner	Heuristic Net
		Multi-phase	Event-driven Process Chain (EPC)
		Organizational Miner	Organizational Model
		Role Hierarchy Miner	Role Hierarchy Model
		Social Network Miner	Similar-Task/ Handover-of-Work/ Subcontracting/ Working Together/ Reassignment
Conformance Checking	To check if reality, as recorded in the log, conforms to the model and vice versa. The comparison shows where the real process deviates from the modeled process.	Transition System Miner	Transition System
		Conformance Checker	Report presenting how much a log matches a model and points of discrepancy
Business Rules Conformance Checking	To check whether some business process is performed as it is intended, i.e. in conformance with certain rules and to get detailed information about the violations of such rules	Graph Matching Analysis, Differences Analysis and Footprint Similarity algorithms	Report presenting how similar are two process models
		LTL Checker; SCIFF Checker	Report presenting if certain properties (business rules) hold in a log

4.2.4 Characterize the “assessable” elements of CMMI-DEV model, identifying which ones are more likely to be examined by Process Mining, in SCAMPI appraisals

In this task, it was identified which elements of the CMMI models are “assessable”, i.e., elements of the CMMI model that are typical target of examinations in SCAMPI appraisals. Criteria were set (as exposed in [31]) for judging the most suitable CMMI “assessable” elements to be investigated by process mining techniques in an appraisal aided by process mining: a) is the implementation of that process done via an organizational standard process? b) is the process execution typically supported by an information system? and c) does the information system export data (when available) as event logs? For this analysis, CMMI-DEV v1.3 process areas and generic practices were considered as “assessable”. As a result, it was found that the most suitable process areas are Causal Analysis and Resolution (CAR) and Requirement Management (REQM) with high suitability. They are followed by Configuration Management (CM), Decision Analysis and Resolution

(DAR), Integrated Project Management (IPM), Organizational Process Focus (OPF), Organizational Training (OT), Product Integration (PI), Project Monitoring and Control (PMC), Project Planning (PP), Process and Product Quality Assurance (PPQA), Quantitative Project Management (QPM), Requirement Development (RD), Technical Solution (TS), Validation (VAL), Verification (VER) with medium suitability. Finally, Measurement and Analysis (MA), Organizational Process Definition (OPD), Organizational Process Measurement (OPM), Organizational Process Performance (OPP), Risk Management (RSKM), Supplier Agreement Management were the ones with low suitability.

Regarding the identification of generic practices (GP), applied criteria were: a) does the practice, as implemented to a particular process area, result in a process with defined and generally sequential activities? and b) is there availability of data in a format expected by process mining? Results were: GP 1.1-Perform Specific Practices, GP 2.1-Establish an Organizational Policy, GP 2.3-Provide Resources, GP 2.4-Assign Responsibility, GP 2.7-Identify and Involve Relevant Stakeholders, GP 2.9-Objectively Evaluate Adherence and GP3.1-Established the Defined Process, with high suitability; followed by GP 2.5-Train People, GP-2.8 Monitor and Control the Process and 2:10 GP-Review Status with Higher Level Management with medium suitability and GP 2.2-Plan the Process, GP2.6-Control Work Products and GP 3.2-Collect Process Related Experiences with low suitability.

4.2.5 Characterize the "assessable" elements in an organization that adopts CMMI, identifying their relationship with Process Mining

In this task, based on the experience of one of the authors, the "assessable" elements in an organization that adopts CMMI were identified. In this scenario, practices (and other elements of the CMMI) serve as a reference for such organizations to define their processes and processes assets (i.e. guides, templates, tools, etc.). In CMMI, they are called "organization's set of standard processes". Once the standard process is defined, each process instance to be executed should adapt the organization's set of standard processes to its specific needs, and after that, such processes (and other assets) should be followed in that instance. In CMMI, the adapted process is called "defined process for the project", even when no tailoring is made. In process mining, this process is a *de Jure* model.

4.2.6 Develop requirements for the proposed method, from findings found in the previous tasks

Eighteen requirements associated with the proposed method were derived during the execution of all previous tasks. Typically, new derived requirements have risen during the conduction of deeper analysis regarding concepts and techniques from both SCAMPI and Process Mining. *The proposed method should clearly identify which part of the reference model scope will be investigated by process mining* is an example of derived requirement for the proposed method which has been raised in the task described in subsection 4.2.4.

4.2.7 Identify alternatives of format (and content) for the proposed method and based on defined criteria, select the best alternative

CMMI models have two published extensions: +SAFE extension to CMMI-DEV v1.2 [34] and Security by Design with CMMI for Development v1.3 [35]. Such extensions were analyzed to identify if there was some standardization among them. It can be said that most of the sections and subsections of existing extensions are common, and so, they were kept in the document that describes the extended method. Other sections and subsections were defined from scratch. Another decision was related to the format regarding the method content. Four options were identified: a) single content: aspects related to process mining are added at any point of the original SCAMPI method document; for instance, in elements such as purpose, implementation guidance, required practices, etc; b) content as extensions: aspects related to process mining are added as "extensions" to the original SCAMPI method; c) content apart: aspects related to process mining are added as isolated processes, activities or practices to the original SCAMPI method, forming a content aside, which is inserted in the document "Process Mining Extension to SCAMPI"; d) content apart with extensions: aspects related to process mining are added or as extensions (for existing content in the original SCAMPI method) or as separated content (for when new processes, activities or practices are needed). It is a hybrid between the b) and c) options. Based on established criteria, there was a tie between the c) and d) options. However, the option d) was chosen because it had a better balance of marks on individual criterion.

4.2.8 Identify methodologies for conducting mining projects and how they can relate to SCAMPI method

For the development of the proposed method, it was assumed that the application of process mining in a SCAMPI appraisal is like a question-oriented process mining project, but with some special characteristics. Therefore, it was necessary to identify the existing methodologies for process mining projects aiming at determining which activities associated with process mining would be added to the existing SCAMPI method. Then, the approach was based on two existing methodologies: PMPM-Process Mining Project Methodology [36] and PM2-Process Mining Project Methodology

[37]. As they were two, it was necessary to integrate them, which resulted in 14 activities distributed in 6 stages, as presented in Table 3. After that, it was identified how the activities of this "integrated approach" could be incorporated in the extended method.

Table 3 – Integration of Process Mining Project Methodologies

Stage	Activity
Scoping and Planning	Identify business processes and associated information systems, and gather basic knowledge
Scoping and Planning	Determine goals and research questions
Scoping and Planning	Determine the required team, data, tools and techniques.
Data understanding	Locate and explore required data in the system's logs
Data understanding	Verify the data in the system's logs and select dataset in terms of event context, timeframe and aspects
Data Processing	Extract the set of required event data
Data Processing	Prepare the extracted dataset, by cleaning, constructing, merging, mapping, formatting and transforming the data
Data Processing	Familiarize and filter log
Process Mining and Analysis	Apply process mining techniques to answer (research) questions
Evaluation	Verify and validate process mining results
Evaluation	Accreditate process mining results
Evaluation	Present process mining results to the organization
Process improvement and support	Identify and implement improvements
Process improvement and support	Support operations

4.2.9 Identify how the identified process mining techniques can be applied as complementary or alternative approaches to data collection and analysis in the SCAMPI method. Identify and develop components to address the requirements

In this task, the requirements were mapped to the components identified in the previous task. It was also identified where, in SCAMPI method, created components should be incorporated and in which format: a) as extension to an existing element; b) as a new process; c) as a new activity or; d) derived from another format (e.g. element "tools and techniques"). Thus, the concept of "Process Mining extension" was created: an element of the extended method, clearly identified, which describes both new processes and/or activities as well as additional considerations of Process Mining when interpreting some other element already present in the standard SCAMPI method. Thus, the extended method, as well as inclusion of new processes and specific process mining activities (using the same format and fields of SCAMPI method, but in gray color), brings extensions to already existing activities in the SCAMPI method in the form of text boxes in gray color at the end of each applicable activity. Table 4 shows locations in the SCAMPI method where the developed components were integrated. Underlined activities are those that received additional content to the original SCAMPI method while processes and activities in bold are new content to the SCAMPI. Note that the titles of new processes and activities received letters rather than (only) numbers to preserve, in the already existing ones, the original numbering from SCAMPI method.

Table 4. Integrating components on original SCAMPI method, reproduced from [31]

Phase	Process	Activities
1 Plan and Prepare for Appraisal	1.1 Analyze Requirements	<u>1.1.1 Determine Appraisal Objectives</u> <u>1.1.2 Determine Data Collection Strategy</u> 1.1.3 Determine Appraisal Constraints <u>1.1.4 Determine Appraisal Scope</u> 1.1.5 Determine Appraisal Outputs 1.1.6 Obtain Commitment to Initial Appraisal Plan

	1.2 Develop Appraisal Plan	<u>1.2.1 Tailor Method</u> <u>1.2.2 Identify Needed Resources</u> <u>1.2.3 Develop Data Collection Plan</u> 1.2.4 Determine Cost and Schedule 1.2.5 Plan and Manage Logistics 1.2.6 Document and Manage Risks 1.2.7 Obtain Commitment to Appraisal Plan
	1.3 Select and Prepare Team	<u>1.3.1 Identify Appraisal Team Leader</u> <u>1.3.2 Select Team Members</u> 1.3.3 Document and Manage Conflicts of Interest <u>1.3.4 Prepare Team</u>
	1.A Obtain Process Mining Artifacts and Elements	1.A.1 Obtain Process Mining Artifacts 1.A.2 Obtain Process Mining Elements
	1.4 Obtain and Inventory Initial Objective Evidence	1.4.1 Obtain Initial Objective Evidence 1.4.2 Inventory Objective Evidence
	1.5 Prepare for Appraisal Conduct	1.5.1 Perform Readiness Review 1.5.2 Re-Plan Data Collection
2 Conduct Appraisal	2.1 Prepare Participants	2.1.1 Conduct Participant Briefing
	2.A Apply Process Mining Techniques on Objective Evidence	2.A.1 Familiarize and Filter Event log 2.A.2 Discover Actual Process from Event Log 2.A.3 Check Conformance of Event Log with de Jure Model 2.A.4 Compare Conformance between de Facto model and de Jure Model 2.A.5 Check Conformance to Business Rules 2.A.6 Examine Process Mining results
	2.2 Examine Objective Evidence	2.2.1 Examine Objective Evidence from Artifacts 2.2.2 Examine Objective Evidence from Affirmations
	2.3 Document Objective Evidence	2.3.1 Take/Review/Tag Notes 2.3.2 Record Presence/Absence of Objective Evidence 2.3.3 Document Model Component Implementation 2.3.4 Review and Update the Data Collection Plan
	2.4 Verify Objective Evidence	2.4.1 Verify Objective Evidence 2.4.2 Characterize Implementation of Model Practices and Generate Preliminary Findings
	2.5 Validate Preliminary Findings	2.5.1 Validate Preliminary Findings
	2.6 Generate Appraisal Results	2.6.1 Derive Findings and Rate Goals 2.6.2 Determine Process Area Ratings 2.6.3 Determine Process Area Profile 2.6.4 Determine Maturity Level 2.6.5 Document Appraisal Results
3 Report Results	3.1 Deliver Appraisal Results	3.1.1 Deliver Final Findings 3.1.2 Conduct Executive Session(s) 3.1.3 Plan for Next Steps

	3.2 Package and Archive Appraisal Assets	3.2.1 Collect Lessons Learned 3.2.2 Generate Appraisal Record 3.2.3 Provide Appraisal Feedback to the CMMI Institute 3.2.4 Archive and/or Dispose of Key Artifacts
4 Action Plan Reappraisal	4.1 Action Plan Reappraisal	4.1.1 Plan Action Plan Reappraisal 4.1.2 Conduct Executive Session(s) Reappraisal 4.1.3 Report Action Plan Reappraisal

It was also necessary to create the content of the (planned) sections in the document "Process Mining Extension to SCAMPI". So, as the main result of the forth research step, the extended method "Process Mining Extension to SCAMPI" was developed. As part of its content, for instance, aspects related to the availability, source and type of data expected by process mining techniques were addressed in the (new) process "1.A Obtain Process Mining Artifacts and Elements" and its derived activities "1.A.1 Obtain Process Mining Artifacts" and "1.A.2 Obtain Process Mining Elements". Details regarding activities under the (new) process "2.A Apply Process Mining Techniques on Objective Evidence" are discussed in section 4.3. Fig. 4 presents an extract of the content of the extended method (in grey), in its definitive format.

1.1 Plan and Prepare for Appraisal

1.1 Analyze Requirements

1.1.1 Determine Appraisal Objectives

Process Mining extension:

After defining appraisal objectives, evaluate which appraisal objectives could be met by applying Process Mining techniques. Later on, such identified objectives will also be used to derive relevant questions that will drive the application of Process Mining techniques in the appraisal. Also, obtained Process Mining results will be later evaluated to check if they meet the objectives defined in this activity.

1.1.2 Determine Data Collection Strategy

Process Mining extension:

Note that due to data-dependency nature of Process Mining, it is only applicable when selected data collection approach is "Verification".

The Data Collection Strategy for an appraisal aided by Process Mining should address the determination of Process Mining specific aspects, such as:

- Description of Process-Aware Information Systems (PAIS) available in the organization unit.
- Derived from the appraisal objectives that can be addressed by Process Mining, the questions that could be answered using Process Mining techniques. See appendix C for examples of questions. Note that Process Mining activities need to be driven by questions. Without concrete questions it will be very difficult to locate, select and extract meaningful event data.
- From identified questions, which Process Mining techniques could be used for collecting and examining Process Mining related artifacts. See appendix C for a list of typical questions.

Fig. 4. Extract from "Process Mining Extension to SCAMPI", reproduced from [31]

4.2.10 Implement and evaluate the method in real situations of software processes assessments. For this evaluation, consider the identified requirements and expectations of usage of the method

The results of this task are presented in the next subsection as well as in subsequent sections.

4.3. Running example

A "running example" was planned to be conducted aiming to test the extended method. The running example refers to data of a software development process, performed in a software house unit in Curitiba, Brazil, which has a strong polo of

software development companies. As noted by Greenfield and Short [38], a software house systematically captures the knowledge of how to produce elements of a specific family of products, making it available in the form of assets such as patterns, frameworks, models and tools, and then applying systematically such assets to automate the development of components, reducing the time and development costs and improving the quality of the final product.

The software development process begins with the demand that is received and recorded in an information system, which also supports other process activities. After that, test scripts are drawn up as well as the specification of the code associated with the demand. After coding it, tests are performed according to the test script, to verify if the code is working. Then, QC-quality control applies a checklist to verify if the specification was correctly coded. If not, conduction of activities regarding test script, coding and testing must be redone. When everything is correct, code and its associated documentation are delivered to customer. Finally, QA-quality assurance performs a verification considering the standards and specifications prior to final environment release.

The focus of the running example was in the new activities (i.e. activities that cover the application of process mining in SCAMPI appraisals) of the extended method (refer to Table 4), and so, the following ones were selected to be conducted: 1.A.1-Obtain Process Mining Artifacts; 1.A.2-Obtain Process Mining Elements; 2.A.1 Familiarize yourself and Filter Event log; 2.A.2 Discover Actual Process from Event Log; 2.A.3-Check Conformance of Event Log with Jure Model; 2.A.4-Compare Conformance between model of Fact and Jure Model; 2.A.5-Check to Conformance Business Rules; 2.A.6-Examine Process Mining Results. Fig. 5 presents the relationship between (new) activities of the extended method and elements associated with process mining and process assessment. In activity 1.A.1, organization provides data to create the *de Jure* model and to identify business rules. Also, data are transformed in event log (via activities 1.A.1 and 2.A.1), which is the basis for generating *de Facto* model, in activity 2.A.2. Then, *de Jure* model as well as business rules are compared with *de Facto* model to identify conformance level (via activities 2.A.3, 2.A.4 and 2.A.5). Finally, in activity 2.A.6 appraisal findings are generated from the conformance checking results.

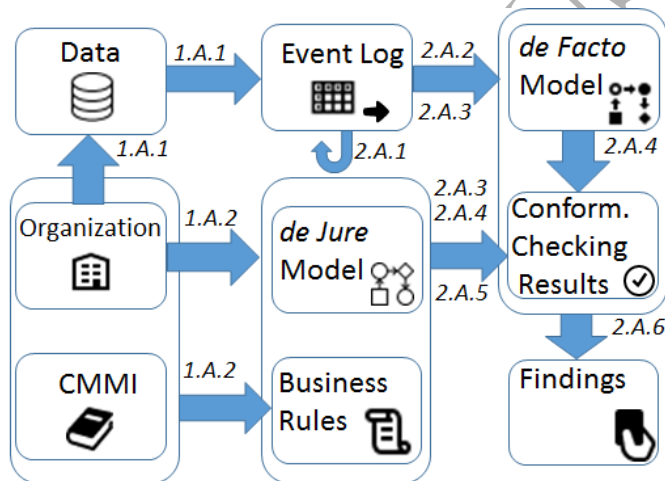


Fig. 5. Relationship between method activities and elements

As result, running example enabled, during and after the development of the method, elaborating and refining the application of process mining techniques in the extended method as well as to test the use of process mining tools such as ProM (www.promtools.org), techniques and algorithms in a real scenario.

5. Method application

The application of the method was done through the conduction of two research steps, 6-Cases study and 7-Experts Review, which are presented in this section. The objective of the cases study was to apply the extended method in real situations in order to perform a verification of the method, while the objective of expert review was to validate it by obtaining a judgment from specialists regarding the usability, feasibility and utility of the proposed method.

5.1. Cases study

In order to identify valid cases (i.e. real situations where the proposed method could be applied), two criteria were established: a) predisposition of organizations undergoing SCAMPI appraisals to conduct them in the manner supported by

process mining and b) existence and availability of data in the format required by (or convertible to) process mining. Two cases were identified as adherent. The cases study comprised the conduction of two SCAMPI C class appraisals using the extended method. They were named as case A-"recommended minimum number of instances" and case B-"all instances". The intention was to use different sampling sizes, so, in case A, the appraisal was conducted using the minimum recommended number of instances (10, as per SCAMPI formula) and in case B, the maximum number of instances of processes available (1911) was used, although the minimum number required by SCAMPI method formula resulted in only 4 instances. The cases refer to two different organizational units of the same information technology company based out of Curitiba, Brazil, although not the same of the running example. Both cases are related to software maintenance operations to their respective customers. Maintenance of the software is carried out via customer service requests using a lifecycle that consists of activities such as "open service request"; "accept / queue service request"; "work in progress"; "pending"; "submit resolution" and "close service request". This lifecycle is supported by software tools, which record data in a format that is readable by process mining tools. Both SCAMPI Class C appraisals aimed to identify the degree of adherence of operations in relation to (selected) CMMI for Services specific (SP) and generic (GP) practices, such as SP 3.1-Receive and Process Service Requests, SP 3.2-Operate the Service System, GP 2.1-Establish an Organizational Policy, GP 3.1-Establish a Defined Process and GP 2.7-Identify and Involve Relevant Stakeholders from Service Delivery (SD) process area. In the cases, all activities in the extended method, including the ones from original SCAMPI, were conducted. It means that appraisal plans were also generated (and executed later on), describing all aspects of the appraisals. Regarding the new activities in the extended methods (see Fig. 4), for instance, in activities 1.A.1-Obtain Process Mining Artifacts and 1.A.2-Obtain Process Mining Elements, an event log was generated from data and also, organizational policies and business rules were identified. *De Facto* process model was obtained, via discovery algorithms, in activity 2.A.2-Discover Actual Process from Event Log. In activity 2.A.3, conformance checking algorithms were applied to identify how much the event log conforms to the *de Jure* model. Conformance checking algorithms were also used to compare conformance between *de Facto* and *de Jure* models in activity 2.A.4-Compare Conformance between *de Facto* model and *de Jure* model. Conformance checking between event log and the identified business rules was performed, using business rules conformance checking algorithms, in activity 2.A.5-Check Conformance to Business Rules. Finally, findings related to select CMMI practices were generated from the examination of process mining results, in activity 2.A.6-Examine Process Mining Results.

Fig. 6 illustrates some screenshots from ProM tool regarding the conduction of case A (although conduction of case B was very similar to case A as well). In the first screenshot there is a result of the application of an algorithm which compared *de Jure* model to *de Facto* model. In the second screenshot, results of the application of an algorithm that performed conformance checking between an event log and its related *de Jure* model. Just as an illustration, an activity highlighted in orange means that there is an inconsistency between the event log and the *de Jure* model. Finally, the third screenshot shows the results of application of an algorithm that performed conformance checking concerning some business rules and the event log.

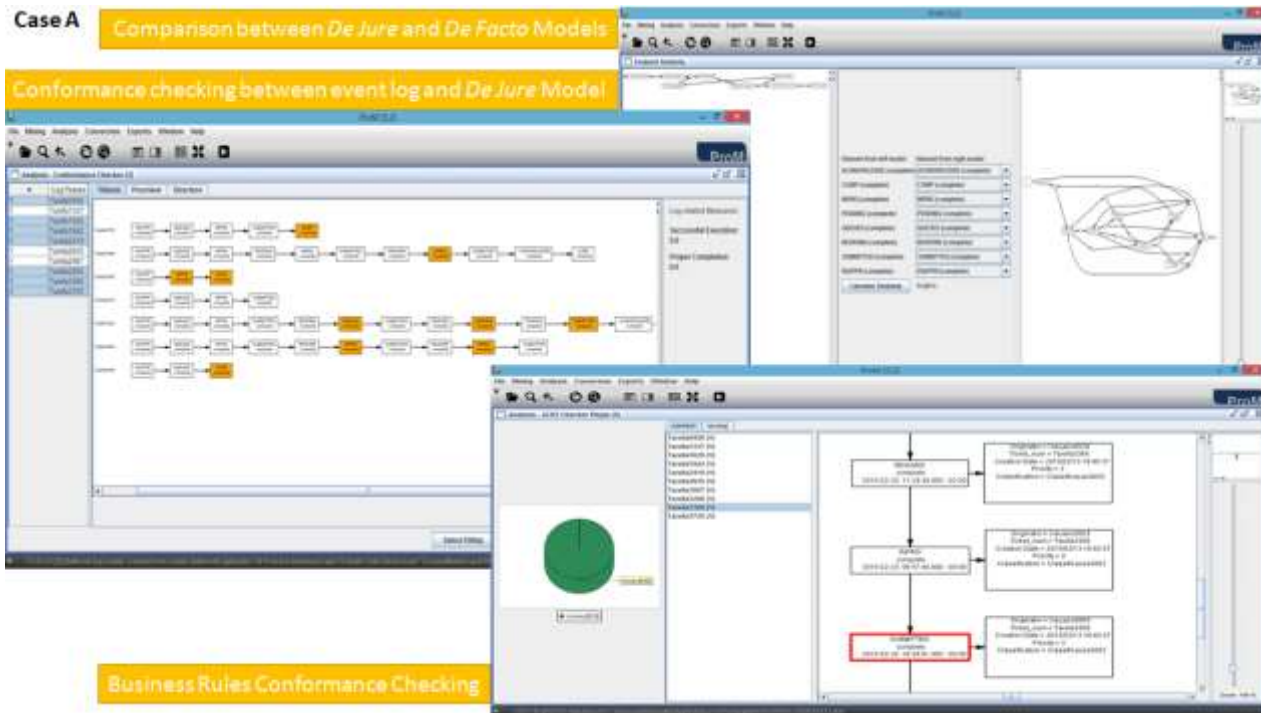


Fig. 6. Screenshots of Case A

In order to evaluate results of cases study, each of the four requirements (refer to Table 1) for the extended method were analyzed, as follows:

- *The extended method should reduce the subjectivity of data analysis and judgment on the implementation of practices.* This requirement has been met since the application of process mining techniques in cases A and B allowed an examination of evidence in a much more objective manner than original SCAMPI method. This is due to the fact that results obtained with the processes mining algorithms are deterministic, i.e., show the same results when repeated with the same parameters and input data. This also allows more confidence for the appraiser when judging the implementation of CMMI practices based on the findings, which were obtained from quantitative information presented as results and indicators by process mining algorithms.
- *The extended method should increase reliability in relation to the selection of the sample, its size and representativeness.* This requirement has been met, since that, with the application of process mining techniques, particularly in case B, it was possible to maximize the sample size. Although the recommended minimum number, as per SCAMPI method, was only 4, in case B, all 1911 process instances available were considered, i.e., more instances than the minimum recommended, making the appraisal results more reliable, in terms of its sampling, when using the extended method.
- *The extended method should reduce the amount of team effort and time (especially in data collection and analysis activities).* This requirement has been met since it was possible to collect and analyze a much larger amount of data (process instances) in much less time than it would take to perform it via traditional data collection and analysis techniques. For example, in case B, all 1911 process instances were examined without affecting the performance of algorithms and time for analyzing results.
- *The extended method should reduce reliance on skills, abilities and experience of appraisers.* This requirement has been met since it was possible to note that, once trained in process mining, member(s) of the appraisal team, playing the role of process mining analyst can apply process mining techniques that provide a) less effort and time for data collection and analysis; b) more objective data analysis and evaluation of implementation of practices and c) higher reliability in relation to sample selection.

As a main result of the cases study, it was judged that method requirements and the appropriateness of usage were met. For example, many more instances were examined in case B when compared with a traditional SCAMPI appraisal at the same circumstances. After the cases study, an expert review was conducted, as presented in next subsection.

5.2. Experts review

In this research step, that cover experts review regarding the proposed method and the results of its application, a conceptual theoretical model was developed to base the questions of the survey. The constructs of such model, derived from identified limitations of SCAMPI method and from objectives for the extended method, are presented in Fig 7. They are:

- q5-objectivity (i.e. is the extended method more objective to analyze data and judge on the implementation of practices, than the original SCAMPI method?);
- q6-reliability (i.e. does the extended method provide more confidence in the selection of the sample and its representativeness, than the original SCAMPI method?);
- q7-independence (i.e. is the extended method less dependent on appraisers and their skills, than the original SCAMPI method?);
- q8-efficiency (i.e. does the extended method demand less team effort and time, than the original SCAMPI method?);
- q9-utility (i.e. is the extended method useful, e.g. minimizes some of the limitations of current SCAMPI method?);
- q10-viability (i.e. are activities, tools and techniques of extended method easy to use?);
- q11-usability (i.e. can the extended method be followed?);
- q12-objectives achievement (i.e. is the extended method feasible, usable and useful defining what, when, where, how and why applying Process Mining techniques SCAMPI appraisals?).

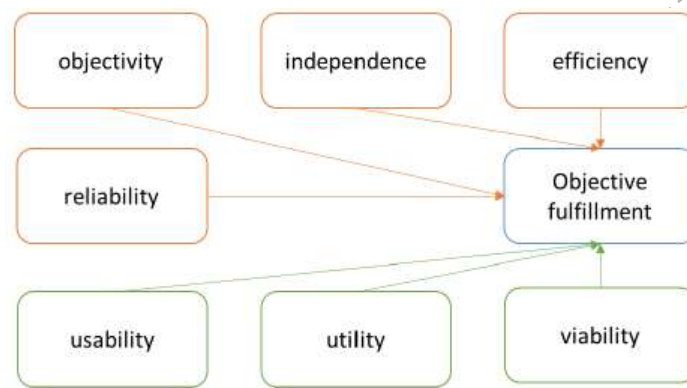


Fig. 7. Model constructs

A questionnaire exploring each one of the constructs was created in Polldaddy tool (www.polldaddy.com). A pilot test was conducted with three experts (a very experienced high maturity lead appraiser, a CMMI expert, and an academic researcher who is also a process mining practitioner), who reflected the target profile of potential respondents. It demonstrated that some questions and related material needed to be adjusted such as the level of details regarding the cases study and the order or content of questions. After adjustments, the survey was made available online in order to reach the largest possible number of SCAMPI appraisers and other CMMI experts, through specific forums of such “community” at LinkedIn (www.linkedin.com).

The intended target for the survey was certified SCAMPI appraisers and other CMMI experts. Concepts of process mining, extracts of the extended method and how the cases study was conducted (as well as its results) were presented to them. For example, Figs. 4 and 6 of this paper were shown to respondents. The survey questions, based on the constructs, were related to the analysis, by experts, whether the extended method reduces the limitations of the traditional SCAMPI method and whether it is viable, useful and usable, as per their opinions. In addition, a walkthrough demonstrating the method, how it was developed and applied, was performed to selected respondents.

The survey was designed using a continuous scale from 1 to 5. It was completed by 50 process improvement experts, although almost all of them without practical experience on process mining. Cronbach's alpha was 0.8981, representing a good internal consistency. The expected sample size was 79. So, 50 respondents would represent, considering a population of 430 units, a confidence interval of 85% (rather than 95%), a sampling error of 10% and 50% of population percentage. With these parameters, the minimum sample size is 49, i.e., less than the number of valid respondents (i.e. 50), making results valid and expandable. For hypothesis testing (one-sample t), it was set as a criterion that for a factor to be considered as satisfied, its value should be higher than 3.00, the midpoint of the range. In hypothesis test ($h_0 = 3$, $\alpha = 0.05$), the results shown in Fig. 8 were obtained, reflecting successful achievement in all factors. The result is also visually presented in Fig. 9, which brings the interval plot for each factor. Note that the confidence interval for the population mean (inferred from the sample) does not contains the value 3.00 in any of the 8 measurements, always being higher, which implies that the

extended method has better performance than the original SCAMPI method, as judged by experts. As an example, q10-viability factor shows a confidence interval between 3.032 and 3.492, i.e., higher than 3.00.

One-Sample T: q5; q6; q7; q8; q9; q10; q11; q12							
Test of $\mu = 3$ vs $\text{not} = 3$							
Variable	N	Mean	StDev	SE Mean	95% CI	T	P
q5	50	3,646	0,694	0,098	(3,448; 3,8429)	6,58	0,000
q6	50	3,712	0,882	0,125	(3,461; 3,963)	5,71	0,000
q7	50	3,402	0,800	0,113	(3,175; 3,630)	3,55	0,001
q8	50	3,605	0,861	0,122	(3,360; 3,849)	4,96	0,000
q9	50	3,732	0,764	0,108	(3,515; 3,949)	6,77	0,000
q10	50	3,262	0,809	0,114	(3,032; 3,492)	2,29	0,026
q11	50	3,652	0,673	0,095	(3,461; 3,844)	6,85	0,000
q12	50	3,648	0,650	0,092	(3,463; 3,833)	7,05	0,000

Fig. 8. Hypothesis testing results

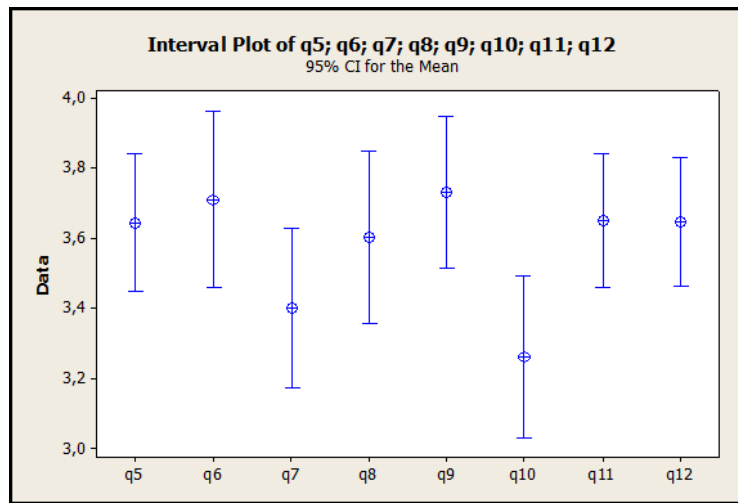


Fig. 9. Interval plot

6. Interpretation and discussion of results

The following section shows the interpretation and discussion of the key results of the research, i.e. the development of the extended method and its application. It includes checking whether research objective is met. Limitations are also presented.

6.1. Development of the extended method

The extended method is the result of exploration and integration of content from disciplines like software and systems engineering; best practice models; process assessment methods and process mining:

- In terms of software and systems engineering, the extended method is the result of performing the following development lifecycle: requirements identification; evaluation of alternatives and decision on the technical solution for the identified requirements; design of the components that address the requirements; implementation of such components; integration of components into the SCAMPI method and verification and validation of the method.
- Concerning best practices models, the extended method is the result of understanding of what are the architectural elements of CMMI, how they are implemented in organizations and how their implementations are judged by SCAMPI appraisers.
- Regarding assessment methods, the extended method is the result of understanding the concepts of artifacts, objective evidence, data collection & analysis strategies and approaches.
- In relation to process mining, the extended method is the result of identification and characterization of process mining techniques, activities and algorithms, as well as the identification of which elements (and how) can be integrated into a specific software process assessment method.

Considered as an essential step for developing the method, the running example helped identifying processes mining tools, techniques and algorithms to be used in the extended method. It was conclusive as to the fact that processes mining techniques can be applied in SCAMPI appraisals, especially a) if there is a log with data for which process mining tools can manipulate and b) if there is identification, in advance, of *de Jure* process and business rules. The running example showed that applying process mining techniques in process assessments also allows for larger sample selection to be examined in an appraisal. Even when sample size is higher, processes mining techniques allow rapid and easy identification, via process discovery, of the actual process being performed. It also allows conformance checking with respect to *de Jure* model through fitness index and other algorithms, such as Footprint Similarity [39] and to business rules, for instance, via Linear Temporal Logic (LTL) Checker [26] or Checker SCIFF algorithm [40]. Thus, appraisals supported by process mining tend to employ less time in evidence collection and analysis. It also assesses more instances than usual. Importantly, this approach implies an extra effort (since new activities need to be performed) during the preparation phase of the appraisal, but that leads to a higher depth and coverage of instances, which makes the sample-based inference - principle of process assessments - more robust while maintaining the same level of effort during the conduction phase of the appraisal when compared with a specific situation of traditional appraisal where a large sample is selected. The running example also served to evaluate the fulfilment of the method requirements (i.e. verification).

6.2. Application of extended method

In relation to the application of the method, the results concern to the conduct of cases study and experts review.

In relation to the cases study, in addition to testing the extended method, it was possible to verify that the extended method is feasible, usable and useful. Also, that process mining techniques can be applied with benefits in SCAMPI appraisals, reducing identified limitations.

In relation to the review by experts, in addition to validating the extended method and its application, it was possible to prove, via hypothesis testing, that the extended method is feasible, usable and useful; and that reduces identified limitations of current SCAMPI method.

Therefore, process mining techniques can be applied with benefits in the SCAMPI appraisal because they reduce limitations as time and effort for data collection and analysis; dependence on experience and competence of appraisers; unreliable sampling and subjectivity on judgment on the implementation of CMMI practices. Thus, the following objective has been achieved: "Develop a feasible, usable and useful method that reduces the limitations of the current SCAMPI method and defines which, when, where, how and why to apply process mining techniques in SCAMPI-based process assessments".

In contrast to the results achieved by the research presented in this paper, there are some limitations associated with the extended method (and its application), such as: a) the extended method covers only one process assessment method (i.e. SCAMPI method); b) due to the nature of process mining techniques, the adequacy of the content of artifacts resulting from the assessed processes are not assessed through the method; c) it does not cover all the process areas of CMMI models or all specific and general practices in the covered process areas; d) the method requires process mining competencies by the appraisers; e) it requires the use of specific process mining tools; f) the extended method requires that process execution data be collected and transformed prior to the conduction phase of the appraisal; g) the method may require regular updates, due to new techniques and process mining algorithms; h) the cases study was limited to very similar cases and was conducted individually by the main author of the proposed method, that can raise some questions regarding its reliability and validity (which are minimized by the fact that process mining techniques are deterministic, i.e. present the same results when repeated with the same parameters and input data); i) measurement and analysis regarding the effort of new activities in the extended method, in comparison to the original SCAMPI method, have not been done; j) the review by experts may not have been potentialized due to the need of achieving a balance between the volume of information to be reviewed by the experts and the level of detail of such information. As a main limitation, it is concluded that since process mining concerns the examination of event logs that are recorded by information systems, performing SCAMPI appraisals aided by process mining are restricted to organizational units that have such characteristics, and also that existing information systems record process-related data in specific format, quality and content as expected by process mining techniques.

7. Conclusion

This section presents the contributions of research, future work, and closing remarks. It also includes checking whether research problem is solved.

Contributions can be categorized in two aspects: Process and Content. In terms of process, the research brings a high level of empirical work and theory test, since it goes beyond theoretical proposition, presenting several research steps

regarding the development and actual application of the proposed method. It is innovative because it applies traditional concepts of software and systems engineering (e.g. V-model and verification & validation practices) in a research context. Regarding content, the research contributes to the current body of knowledge of process assessments by adding a usable, useful and feasible method; that reduces some limitations of the SCAMPI method and constitutes a clear and novel guide of what, how, where, when and why to apply process mining techniques in SCAMPI appraisals. The extended method position process mining as a complementary approach (and substitute at some points) to the traditional SCAMPI data collection & analysis techniques but also for judging on the implementation of CMMI practices during an appraisal.

Future work on the current topic are therefore recommended:

- Development of studies regarding current SCAMPI method in order to enable the judgment of CMMI practices based on indicators or other quantitative criteria;
- Improvement of the extended method aiming to guide on coverage of specific practices and/or process areas of CMMI for Services model;
- Evolution of the current focus of the extended method (i.e. conformance) for a method to also examine performance and process improvement aspects;
- Application of the method extended into new SCAMPI appraisal scenarios (e.g. SCAMPI B and A), to corroborate the findings presented here and allow other generalizations;
- Application of the extended method in similar scenarios, such as internal and external audits;
- Automation of process mining techniques in SCAMPI appraisals through workflow automation tools such as RapidMiner (www.rapidminer.com). This tool already has an extension called RapidProm [41] that allows to automate various process mining algorithms, including some of the algorithms proposed in the extended method;
- Expanding the scope of extended method to other process assessments methods.
- Development of similar methods for application of process mining techniques in process improvement methods such as Lean or Six Sigma.

Finally, it can be said that the research problem was solved since there is not anymore a lack of a method that defines what, when, where, how and why to apply process mining techniques in software process appraisals, aiming to reduce the limitations of the current SCAMPI method. The results also indicate that applying process mining techniques in SCAMPI appraisals enhances how the appraisal team collects and analyzes data as well as judges the extent to which an organizational unit has implemented the reference model practices being appraised. Appraisers and process analysts can now rely on a guidance regarding identifying which process mining algorithms to use and how to apply them to answer the typical "questions" of software process assessments (as presented in [31]).

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